

Apple Developer & TestFlight — Step by Step

Everything needed to get Gap Sense onto participants’ iPhones with working vibration, in the order to do it.

Before anything: you may not need to pay at all

Ask Professor Hong whether Stevens already has an Apple Developer Program membership. Many universities do, often run by IT or a research computing group.

If Stevens has one, an Account Holder or Admin can add you to the team in about two minutes, at **no cost**. You would get a “Developer” or “App Manager” role, which is enough to upload builds and run TestFlight.

This is worth one email before spending anything. It skips the \$99, skips the 1–2 day enrolment wait, and puts the app under the university rather than your personal name — which is better for a research project that might later be published or handed on.

If Stevens does not have one, there are still two routes:

Route	Cost	Speed	App belongs to
Individual enrolment	\$99/year	Fastest — often approved in a day	You personally
Organization enrolment (Stevens)	Free for accredited educational institutions	Slower — needs a D-U-N-S number and verification	Stevens

Apple waives the fee for accredited educational institutions enrolling as an organization. Stevens qualifies. But it needs someone with legal authority to bind the university to sign, plus a D-U-N-S lookup, so it typically takes weeks rather than days. Only worth it if the conference is far enough away.

Do I have to pay right now?

Short answer: you can start now for free, but you cannot finish without paying.

Here is exactly where the wall is:

Step	Cost	Can you do it before paying?
Create an Apple ID	Free	✓ Yes
Create a free Apple Developer account	Free	✓ Yes
Install Xcode, open the project	Free	✓ Yes
Build and run on your own iPhone	Free	✓ Yes — but the app stops working after 7 days
Enrol in the Apple Developer Program	\$99/year	✗ Payment is taken at enrolment. It cannot be deferred.
Upload to App Store Connect	Included	✗ Needs the paid membership
TestFlight link for participants	Included	✗ Needs the paid membership

So: **do steps 1–4 today at no cost.** That gets the app onto your own iPhone so you can test the vibration yourself, which is the thing you most need to verify anyway. Then get approval for the \$99, and the rest takes about an hour.

There is no trial, no invoice-later, and no student discount on the Program fee.

Something to send Professor Hong

Feel free to copy this.

To let conference participants test the vibration feedback on iPhone, we need an Apple Developer Program membership — \$99/year. There is no alternative: Safari on iOS cannot vibrate at all, so haptics require a real installed app, and TestFlight is Apple's only route for distributing test builds to people outside our team.

This matters specifically for DeafBlind participants, for whom vibration is the only feedback channel available — they can use neither the audio nor the visual output.

Two questions before we buy anything:

1. Does Stevens already have an Apple Developer Program membership I could be added to? That would cost nothing.
2. If not, should we enrol as an individual (\$99/year, approved in about a day, registered under my name), or as Stevens (fee waived for accredited educational institutions, but needs a D-U-N-S number and someone authorised to sign, so several weeks)?

Given the conference timing, the individual route is the practical one unless Stevens already has an account.

Route A — Individual enrolment (the fast one)

Step 1. Apple ID with two-factor authentication — free

Use an Apple ID you control long term. Your Stevens address is a good choice if the account should outlive your studies.

Two-factor authentication is **required**. Turn it on first at appleid.apple.com → Sign-In and Security.

Step 2. Free developer account — free

Sign in at developer.apple.com/account with that Apple ID and accept the agreement. This costs nothing and takes a minute.

You can now open the project in Xcode, sign it with your “Personal Team”, and install it on your own iPhone. **Do this now** — it lets you feel the haptics yourself before spending anything. The app will stop opening after 7 days; just re-run it from Xcode.

Step 3. Enrol in the Program — \$99/year

At developer.apple.com/programs/enroll:

- Choose **Individual / Sole Proprietor** (not Organization — that needs a D-U-N-S number)
- Your **legal name must match** your payment card and any ID Apple asks for
- Pay \$99

Apple may ask to verify your identity. Approval is usually **24–48 hours**, sometimes same day.

If you are not a US citizen or permanent resident, Apple may request extra identity documents. Allow a little more time.

Step 4. Create the app record — after approval

At appstoreconnect.apple.com → **Apps** → + → **New App**:

Field	Value
Platform	iOS
Name	Gap Sense
Primary language	English (U.S.)
Bundle ID	com.gapsense.crossingpractice
SKU	gapsense-001 (any unique string)
User access	Full Access

The bundle ID is already configured in the project, so it should appear in the dropdown. If it does not, create it first under **Certificates, Identifiers & Profiles** → **Identifiers**.

Step 5. Upload a build

```
cd ios-app
npx cap sync ios
```

```
npx cap open ios
```

In Xcode:

1. Select the **App** target → **Signing & Capabilities** → set **Team** to your new membership
2. Set the device selector at the top to **Any iOS Device (arm64)** — you cannot archive while a simulator is selected
3. **Product** → **Archive**
4. When the Organizer opens: **Distribute App** → **TestFlight & App Store** → **Upload**

Processing takes 5–15 minutes. You will get an email when it is ready.

Export compliance is already answered in the project (`ITSApplUsesNonExemptEncryption` is set to `NO`), so Apple will not ask you about encryption on every upload.

Step 6. Beta App Review — once

In App Store Connect → your app → **TestFlight**:

1. Fill in **Test Information**: what to test, and a contact email
2. Add an **external** testing group — call it something like “Conference”
3. Add the build to that group and **Submit for Review**

Beta App Review usually takes 1–2 days, and is only needed for the first build. Later builds go out immediately.

What to write in “What to Test”:

Gap Sense helps blind and low-vision travellers build an intuitive sense of how long their street crossing takes, using sound and vibration rather than numbers of seconds. Please test with VoiceOver on. Enable “Vibrate on cue” under Accessibility and confirm the vibration patterns are distinguishable. No account or login is needed; the app collects no data.

Step 7. Get the public link

Once approved: **TestFlight** → **your group** → **Public Link** → **Enable**.

You get a URL anyone can use, with **no need to collect emails in advance**. Up to **10,000 testers**.

Email that link to participants. They tap it, install TestFlight if they do not already have it, and tap Install. About 90 seconds.

Copy-paste pack

Everything you will be asked to type, written out so you are not composing text at the keyboard while a form waits.

App record (App Store Connect → Apps → +)

Field	Value
Platforms	iOS only
Name	Gap Sense
Primary Language	English (U.S.)
Bundle ID	com.gapsense.crossingpractice
SKU	gapsense-001
User Access	Full Access

Beta App Description

Gap Sense is an Orientation & Mobility training tool for blind and low-vision travellers. It builds an intuitive, felt sense of how long it takes the user to cross a street, so they can judge whether a gap in traffic is long enough.

Feedback is given as duration – sounds, visual flashes and vibration – never as a number of seconds. The app has four activities: measuring your crossing, practising

the timing, comparing a gap against your crossing time, and doing the same judgement with real traffic.

Feedback Email

ahaque3@stevens.edu

What to Test

Please test with VoiceOver enabled.

1. Open Accessibility and turn on "Vibrate on cue", then press "Test haptic pulse". You should feel three clearly different sensations: a single tap, a light triple, and one long heavy buzz. Please tell us if they are hard to tell apart, especially through a pocket or glove.
2. From the home screen, work through the four activities in order. Each one is numbered.
3. In "Practise my timing", listen for the gap between your own tap and the reference sound. That gap is the feedback. Please tell us if you found yourself counting seconds instead of feeling the duration.
4. Try the app outdoors near traffic if you can, and tell us whether the two feedback sounds are audible and clearly different from each other.

No account or login is needed. The app collects no data and makes no network calls. Everything stays on your device.

Export compliance

Already answered in the project (`ITSAppUsesNonExemptEncryption` = NO). If a form asks anyway: **No**, the app does not use encryption beyond standard HTTPS.

Demo account

Not required — the app has no login.

While you wait for activation

The membership only gates TestFlight. Everything below works today with a free Apple account, and answers questions that waiting will not.

1. Put the app on your own iPhone (free, 7-day expiry)

```
cd ~/Desktop/streets-main/ios-app
npx cap sync ios
npx cap open ios
```

Xcode → Settings → Accounts → add your Apple ID. Select your iPhone in the device dropdown. Press ⌘R. First launch fails with “Untrusted Developer” — on the phone go to Settings → General → VPN & Device Management → your Apple ID → Trust, then open the app.

2. Test the haptics. Accessibility → “Vibrate on cue” → Test haptic pulse. Three distinct sensations? Try it through a trouser pocket and with the phone held loosely, not just flat on a palm.

3. Test the sounds outdoors. Stand near real traffic. Are the within-margin chime and the outside-margin buzz each audible, and clearly different?

4. Test with VoiceOver on. Run a full practice. Note especially whether the double-tap-to-activate delay throws off your marking accuracy.

5. Test the Android build. `ios-app/build/GapSense-android-debug.apk` is signed and installs by sideload. Android vibrates in the browser too, so you can compare the native and web haptics directly.

Anything that comes out of these is far more useful than more waiting.

Timeline

	Time	Cost
Apple ID + free developer account	10 minutes	Free
Test on your own iPhone via Xcode	30 minutes	Free
Program enrolment approval	1–2 days	\$99
App record + upload	1 hour	—
Beta App Review	1–2 days	—
Total before you have a shareable link	~4–5 days	\$99

The waiting is the bottleneck, not the work. If the conference is within two weeks, start the enrolment as soon as you have approval to spend.

Things that will otherwise catch you out

TestFlight builds expire after 90 days. Upload a fresh one close to the conference.

Increment the build number every upload. App Store Connect rejects duplicates. In Xcode: target → General → Build. Any increasing number works.

Archive requires a real device target. With a simulator selected, the Archive menu item is greyed out.

Free-account builds last 7 days. That is the personal-team limit, not a problem with the app. Re-run from Xcode to refresh.

The simulator cannot do haptics. There is no Taptic Engine to drive. Test vibration on a real iPhone only.

Checklist

Today, free: - ☐ Ask Professor Hong about an existing Stevens Apple Developer account - ☐ Apple ID with two-factor authentication - ☐ Free developer account at developer.apple.com - ☐ Install the app on your own iPhone from Xcode and **feel the haptics yourself**

Once you have approval to spend: - ☐ Enrol as Individual, \$99 - ☐ Wait for approval (1–2 days) - ☐ Create the app record with bundle ID `com.gapsense.crossingpractice` - ☐ Archive and upload from Xcode - ☐ Submit for Beta App Review (1–2 days) - ☐ Enable the public TestFlight link - ☐ Test that link yourself on a phone that has never had the app

Before the conference: - ☐ Fresh build uploaded (90-day expiry) - ☐ Both links in the invitation email — the web link to try it, the TestFlight link to feel it